

Bus Ticket Booking System - Project Documentation

Project Overview

The Bus Ticket Booking System is a console-based Java application that allows customers to register, log in, view available buses, book tickets, view booked tickets, cancel tickets, and view bus summaries. The system maintains information about buses, customers, seats, and tickets.

Objectives

- Customer registration and authentication - Display available buses - Seat booking and cancellation - Ticket management - Bus occupancy tracking

Modules

1. Customer Management 2. Bus Management 3. Ticket Booking 4. Ticket Cancellation 5. Reporting and Summary

Class Responsibilities

Bus Class

Stores bus details such as bus ID, bus type (AC/Non-AC), seat type (Seater/Sleeper), total seats, and seat availability. Responsible for seat booking, seat cancellation, and displaying available seats.

Customer Class

Stores customer information such as customer ID, name, password, age, and gender. Used for authentication and maintaining user details.

Ticket Class

Stores ticket information including ticket ID, bus details, booked seats, fare amount, customer ID, and number of tickets booked.

BusTicketBookingSystem Class

Acts as the central controller of the application. It manages customers, buses, tickets, login sessions, booking operations, cancellation operations, and reports.

Features

User Registration

Customers can create an account by providing name, password, age, and gender. A unique customer ID is generated automatically.

Login

Registered users can log in using their customer ID and password.

View Available Buses

Displays all buses with their type, seat category, total seats, and available seats.

Book Ticket

Customers select a bus, choose the number of tickets, and reserve available seats. The system validates seat availability before confirming the booking.

Fare Calculation

Fare is calculated based on bus type and seat type. AC buses have a higher fare than Non-AC buses, and Sleeper buses include an additional charge.

Cancel Ticket

Customers can cancel booked tickets. Seats become available again and the fare amount is displayed as the refund value.

Bus Summary

Displays occupancy details including total seats, booked seats, and available seats for each bus.

Future Enhancements

- Route and destination management - Dynamic pricing - Online payment integration - Waiting list management - Admin dashboard - Database integration using MySQL - Email/SMS notifications